Sparse Autoencoders for Matrix-Valued Recurrent States

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Abstract

Reconstruction MSE and downstream fidelity diverge for sparse autoencoders trained on GatedDeltaNet recurrent states: a bilinear encoder ($\mathbf{v}_i^\top S_t\, \mathbf{w}_i$) matched to the rank-1 write geometry of the 128×128 state underperforms a flat baseline on MSE yet yields lower perplexity under full state replacement ($+1.33\%$ vs. $+3.31\%$ at layer 9), with $64\times$ fewer parameters and half the FLOPs. A 720-run encoder-swap ablation locates the advantage in the encoder: pairing the bilinear encoder with a flat decoder drops perplexity to $+0.58\%$, even though flat SAEs win MSE on every layer. The dissociation traces to residual coherence. Flat residuals are $5.2\times$ more correlated across heads than bilinear residuals and compound when 16 heads are replaced jointly. The spectral ratio σ_1/σ_2 predicts where rank-1 decoding helps (Spearman ρ=0.69, p=0.013, N=12 layers) and transfers to DeltaNet 1.3B (ρ=0.58, p=0.006, pooled across 21 layers and both architectures).

1 Introduction

Hybrid language models route most of their computation through recurrent layers. In Qwen3.5 [Qwen Team, 2026], ablating the 18 GatedDeltaNet layers causes a $35{,}200\times$ perplexity increase; ablating the 6 attention layers causes $82\times$ [Borobia et al., 2026]. Each GatedDeltaNet layer [Yang et al., 2024] carries 16 heads, each with a 128×128 state matrix S_t updated by a gated rank-1 outer product:

$$S_t = \alpha_t \odot S_{t-1} + \beta_t\, \mathbf{k}_t \mathbf{v}_t^\top. \tag{1}$$

SAEs decompose activations into sparse, interpretable features [Bricken et al., 2023, Elhage et al., 2022, Gao et al., 2024]. Recent work extends SAEs beyond the residual stream to attention outputs [Kissane et al., 2024] and evaluates dictionaries by intervention quality, erasure behavior, or benchmark suites rather than reconstruction alone [Karvonen et al., 2024, 2025, Dunefsky et al., 2025]. Recurrent-interpretability studies probe Mamba-style models and hidden states [Gu and Dao, 2023, Ali et al., 2024, Sharma et al., 2024, Paulo et al., 2024, Hossain et al., 2025], but to our knowledge, sparse factorization has not been applied to the matrix-valued recurrent state itself. The standard approach, vectorizing a 128×128 state into $\mathbb{R}^{16384}$, discards the rank-1 write geometry. We design SAE variants that exploit this geometry and test three claims:

1. **Encoder drives fidelity.** Flat SAEs achieve lower MSE on all tested layers, yet a bilinear encoder that scores atoms via $\mathbf{v}_i^\top S_t\, \mathbf{w}_i$ yields lower downstream perplexity. A 720-run encoder-swap ablation isolates the encoder: bilinear encoder + flat decoder gives $+0.58\%$ perplexity at layer 9 vs. $+1.33\%$ for bilinear and $+3.31\%$ for flat.
2. **Residual coherence.** The MSE-fidelity gap traces to error structure: flat residuals are $5.2\times$ more correlated across heads and compound under joint replacement, while bilinear residuals stay independent.
3. **Spectral prediction.** σ_1/σ_2 predicts where a rank-1 decoder beats a flat baseline (Spearman ρ=0.69, p=0.013, N=12 layers) and transfers to DeltaNet 1.3B (ρ=0.58, p=0.006, 21 layers pooled).

At matched dictionary size, the bilinear encoder uses $64\times$ fewer parameters, half the FLOPs, and achieves lower downstream perplexity than flat.

Preprint.

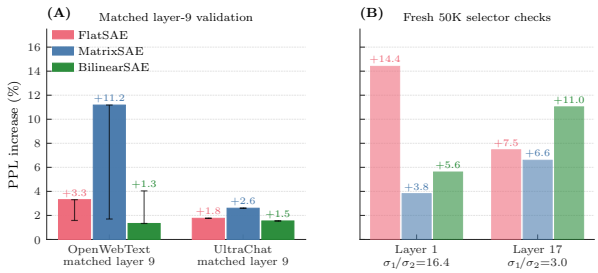

Figure 1: Perplexity increase under full state replacement (Qwen3.5-0.8B, n_f=2048, k=32). A bilinear encoder matched to the rank-1 write geometry adds only +1.33% perplexity, vs. +3.31% for a flat baseline and +11.18% for a rank-1 decoder paired with a flat encoder. **Left:** Per-head matched cohort, layer 9 (n=16 heads); bar heights are the canonical seed-42 checkpoint used throughout the paper, error bars span the observed range (min–max) across seeds $\{42, 0, 1\}$, not a statistical confidence interval. **Right:** Cross-layer comparison from a separate 50K-trained cohort; high-ratio layers favor rank-1 structure.

2 Method

Model and data. Qwen3.5-0.8B contains 18 GatedDeltaNet layers and 6 attention layers across 24 total blocks. Each GatedDeltaNet layer has 16 heads, each maintaining a 128×128 state matrix S_t updated by Equation 1. We extract final-token states from 5,000 OpenWebText sequences (length 1024), split 80/20 into train and validation sets. The spectral audit (Section 3) samples 12 of the 18 GatedDeltaNet layers; the main SAE sweep focuses on layers 1, 9, and 17 because they span concentrated to diffuse spectra. Downstream evaluation (Section 5) uses a per-head matched cohort at layer 9. The encoder-swap ablation uses the same 0.8B state extraction pipeline. It trains five architectures per head with three seeds each across layers 1, 9, and 17 (720 runs total), then averages over all 16 heads per layer. Causal steering (Appendix A) runs at a different model scale (Qwen3.5-4B) with its own prompt set. A feature is *alive* if it receives a nonzero TopK activation on at least one validation sample; dead otherwise.

Architectures. The state S_t accumulates rank-1 outer products $\mathbf{k}_t\mathbf{v}_t^\top$ through gated updates. An unconstrained dictionary atom in $\mathbb{R}^{128\times128}$ has 16,384 free parameters; a rank-1 atom $\mathbf{v}_i\mathbf{w}_i^\top$ has 256 and matches the update geometry exactly (J. Batson, personal communication, 2026). All three architectures (Table 1) learn a sparse code over n_f atoms with TopK selection (k=32, selecting the 32 highest-scoring atoms per input). Let $\mathbf{x} = \mathrm{vec}(S_t - M)$, where M is the per-head mean state. FlatSAE applies a dense encoder-decoder to $\mathbf{x}$ and ignores matrix geometry. MatrixSAE keeps the same linear encoder but constrains each decoder atom to a rank-1 outer product. BilinearSAE replaces the dense encoder with a bilinear matched-filter score $\mathbf{v}_i^\top S_t\, \mathbf{w}_i$ and keeps the rank-1 decoder. The encoder-swap ablation isolates encoder and decoder effects under matched data, dictionary size, sparsity, optimizer, and training budget. The five variants differ only in factorization: FlatSAE (dense encoder, dense decoder), MatrixSAE (dense encoder, rank-1 decoder), BilinearSAE (bilinear encoder, rank-1 decoder), bilinear-flat (bilinear encoder, dense decoder), and tied bilinear (bilinear encoder sharing decoder weights, rank-1 decoder). We report layer-averaged validation MSE across the 16 heads in each layer, then compare that ordering with the downstream replacement results. BilinearSAE uses 0.50× the FLOPs of FlatSAE per sample (33.6M vs. 67.1M), trains ∼3× faster per epoch, and uses 64× fewer parameters.

Training. All SAEs minimize MSE plus an auxiliary dead-feature loss that reconstructs residuals through inactive atoms. We use Adam with a cosine learning rate schedule: linear warmup over 50 steps, peak 3×10^{-4}, decaying to 3×10^{-5}. Batch size is 256; training runs 20 epochs. We resample dead features (inactive for 100 consecutive steps) every 250 steps by reinitializing encoder and decoder directions from high-loss training samples. We re-normalize decoder columns every 100 steps. Each configuration runs with 3 seeds for error bars. All SAE training runs on a single NVIDIA A10G (24 GB); the full 300+ experiment sweep takes ∼48 GPU-hours on Modal. Table 1 shows parameter counts at the reference dictionary size n_f=16,384 (1× expansion over the 128×128 state); the matched-cohort evaluation (Sections 3–5) uses n_f=2,048 with k=32. Sweeps over $n_f \in \{1024, 2048, 4096, 8192\}$ appear in Appendix D.

3 Spectral Prediction

Spectral concentration across layers. Before training any SAE, we check how close GatedDeltaNet states sit to rank-1. Across 12 sampled layers, spectral concentration varies widely: the

Table 1: Architecture comparison at n_f=16,384 (1× expansion). Replacing the dense encoder with n_f pairs of 128-d vectors cuts parameters 64× from FlatSAE to BilinearSAE.

	FlatSAE	MatrixSAE	BilinearSAE
Encoder	$W_e\,\mathrm{vec}(S_t)$	$W_e\,\mathrm{vec}(S_t)$	$\mathbf{v}_i^\top S_t\,\mathbf{w}_i$
Decoder	$W_d\,\mathbf{a}$	$\sum a_i\,\mathbf{v}_i\mathbf{w}_i^\top$	$\sum a_i\,\mathbf{v}_i\mathbf{w}_i^\top$
Params	537M	272M	8.4M

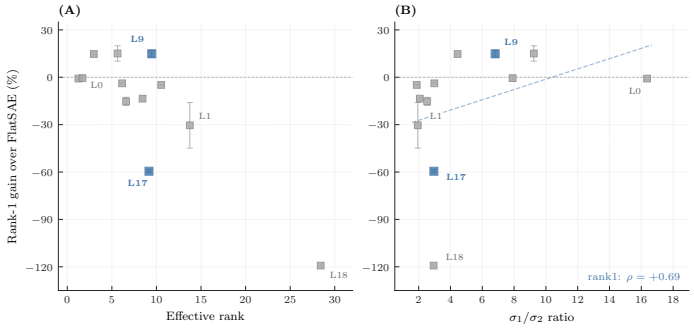

Figure 2: σ_1/σ_2 helps identify when rank-1 decoding beats the flat baseline ($\rho = 0.69$, $p = 0.013$; N=12 sampled GatedDeltaNet layers, Qwen3.5-0.8B, 4,000 training states). **Left:** Layers 9 and 17 have similar effective rank but different spectral peaks. **Right:** σ_1/σ_2 separates winners from losers; error bars are ±1 SEM over 3 seeds.

earliest layer has effective rank 1.25 (σ_1/σ_2=16.4), the most diffuse reaches effective rank 28.4. Effective rank is a weaker predictor of decoder preference. Layers 9 and 17 have similar effective rank, but layer 9 has a clear spectral peak (σ_1/σ_2=6.81) while layer 17 is flat (σ_1/σ_2=2.96). MatrixSAE has 14.9% lower validation MSE than FlatSAE at layer 9 but 59.5% higher at layer 17.

The ratio σ_1/σ_2 separates these cases and serves as a zero-shot diagnostic (Figure 2). Across 12 sampled layers, Spearman ρ=0.69 (p=0.013): layers with concentrated spectra favor MatrixSAE, while flatter spectra eliminate the MSE gap. Leave-one-out jackknife keeps ρ between 0.60 and 0.74 (11/12 folds at p<0.05; bootstrap 95% CI $[0.21, 0.90]$). This diagnostic requires one SVD per head on a small state sample, no SAE training; the ratio is stable across sample sizes (CV $<$ 3% at N=100, $<$ 1% at N=1,000).

Transfer across models. DeltaNet uses a different update rule from GatedDeltaNet but maintains the same 128×128 state shape. We evaluate `fla-hub/delta_net-1.3B-100B` (24 layers, 16 heads, $(128, 128)$ states) on 5,000 samples with 3 seeds per SAE type. BilinearSAE wins by 49% on the high-ratio layer (σ_1/σ_2=11.3) and loses on both low-ratio layers (1.5, 5.2), matching the GDN pattern. Pooling all 21 layers across both architectures gives Spearman ρ=0.58, p=0.006; below σ_1/σ_2=4, flat achieves lower MSE in all 13 cases. At Qwen3.5-4B (24 GatedDeltaNet layers, 32 heads, $(128, 128)$ states), bilinear adds +0.59% downstream perplexity at layer 9 vs. flat's +1.08%, and MatrixSAE's penalty drops from +11.18% at 0.8B to −0.01% at 4B, consistent with more concentrated spectra at scale. The predictor identifies *when* rank-1 structure helps, not the optimal decoder rank: rank-2 decoding at layer 17 cuts MSE by 29% but worsens downstream perplexity (+2.35% vs. +1.17% for rank-1).

4 Write Alignment

The BilinearSAE dictionary is trained to reconstruct S_t, not to match the individual $\mathbf{k}_t\mathbf{v}_t^\top$ writes that produced it. For each alive feature, we measure alignment between its decoder atom $\mathbf{v}_i\mathbf{w}_i^\top$ and the GatedDeltaNet write vectors at positions where the feature fires, using a combined score $\sqrt{|\cos_k||\cos_v|}$ over the top 50 activating positions. Mean combined alignment is 0.137 (1.9× the random null of 0.071), and 17.5% of features exceed 0.3 against 0% for random rank-1 atoms across 1,000 null trials (Figure 3). Partial alignment, despite no write-level supervision, indicates the rank-1 constraint channels the dictionary toward the model's write geometry. Per-feature ablation shows no single alignment percentile carries disproportionate downstream weight ($\leq$0.02 pp across top/bottom/random 20% groups): the advantage is distributed across the dictionary, not concentrated in a few high-alignment atoms.

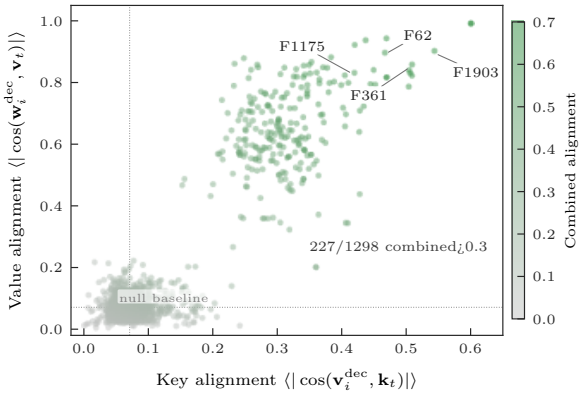

Figure 3: SAE decoder atoms recover actual GatedDeltaNet write vectors (BilinearSAE, layer 9, head 4, Qwen3.5-0.8B, n=1,298 alive features). Each point is one alive feature; axes show mean $|\cos|$ between the decoder atom and the write vectors $(\mathbf{k}_t, \mathbf{v}_t)$ at positions where the feature fires. Dotted lines mark the random-vector axis baseline $E[|\cos|] \approx 0.071$; the colorbar shows the geometric-mean combined alignment $\sqrt{|\cos_k||\cos_v|}$. Mean combined alignment across all 1,298 alive features is 0.137 (null: 0.071); 227 features (17.5%) exceed the illustrative 0.3 threshold, vs. 0% for random rank-1 atoms across 1,000 null trials.

5 Downstream Fidelity

Structural alignment between decoder atoms and write vectors (Section 4) does not by itself guarantee functional fidelity. Under teacher-forced evaluation on held-out OpenWebText, we replace all 16 layer-9 states with their SAE reconstructions before each next-token prediction and measure perplexity against gold continuations.

The bilinear SAE adds only +1.33% perplexity on the matched layer-9 cohort (Figure 1). FlatSAE costs +3.31%, and MatrixSAE costs +11.18%, an 8× larger penalty than bilinear despite sharing the same rank-1 decoder. The ordering transfers to UltraChat (Appendix J) and across layers: at layer 1 (σ_1/σ_2=16.4), MatrixSAE incurs +3.81% vs. FlatSAE's +14.40%; at layer 17 (σ_1/σ_2=2.96), the gap narrows to $\leq$1 pp.

A parameter-matched FlatSAE (n_f=256, 8.4M parameters) incurs +7.36% perplexity, 5.5× worse than bilinear at matched parameter count. A rank-r SVD baseline ($r \in \{1, 2, 4, 8, 16, 32\}$) produces $\leq$0.1% perplexity change at every rank: the downstream cost is from TopK sparsity, not dimensionality reduction.

Encoder-swap ablation. To isolate which component drives the gap, we train five SAE variants on all 16 heads across layers 1 ($\sigma_1/\sigma_2 \approx 12$), 9 (6.8), and 17 (3.0): flat, rank-1, bilinear, bilinear-flat, and tied bilinear (5 types × 16 heads × 3 layers × 3 seeds = 720 runs). Bilinear-flat (bilinear encoder, dense decoder) achieves +0.58% perplexity at layer 9, below BilinearSAE (+1.33%) and FlatSAE (+3.31%). The full ordering is bilinear-flat (+0.58%) < bilinear (+1.33%) < flat (+3.31%) < tied bilinear (+7.05%) < rank-1 (+11.18%): the top two variants share a bilinear encoder with independent projections. Swapping between rank-1 and dense decoders (bilinear vs. bilinear-flat) shifts perplexity by 0.75 pp; swapping between bilinear and dense encoders (bilinear-flat vs. flat) shifts it by 2.73 pp. Bilinear-flat uses 34.6M parameters (bilinear encoder 1.05M + dense decoder 33.6M), fewer than flat's 67.1M, so the advantage is not from decoder capacity. The spectral predictor tracks the encoder benefit: at layer 17, bilinear-flat gives +0.46%; at layer 1, +2.92%. Tied bilinear costs +7.05% (mean encoder-decoder cosine: 0.25 ± 0.44), confirming that encoder and decoder directions need to diverge. Replacing one head at a time shows equal loss deltas for bilinear and flat (paired t-test: p=0.50, n=16); the advantage appears only when all 16 heads are replaced jointly.

Residual coherence. The per-head parity but joint-replacement gap points to error structure. With $\mathbf{r}_h \in \mathbb{R}^{16384}$ the per-token head-h residual on held-out data, the coherence ratio $C_k = \|\sum_{h=1}^{k} \mathbf{r}_h\| / \sqrt{\sum_{h=1}^{k} \|\mathbf{r}_h\|^2}$ equals 1 for independent residuals and $\sqrt{k}$ under perfect alignment. Flat errors are 5.2× more correlated across heads than bilinear's (mean pairwise $|\cos|$=0.026 vs. 0.005), and C_{16} reaches 1.079 for flat vs. 1.001 for bilinear. The source: flat SAEs reuse the same alive atoms across heads (Jaccard overlap 0.79 vs. 0.29), creating a shared null space whose residuals point in similar directions and compound. Per-head MSE favors flat (3.2×10^{-5} vs. 3.7×10^{-5}; flat wins layer-average MSE on all three layers, p=0.057–0.0002 paired t-tests, n=16; Table 9, Appendix K). The downstream gap is from correlated residuals, not reconstruction magnitude.

6 Related Work

SAEs decompose model activations into sparse features [Cunningham et al., 2023, Bricken et al., 2023, Templeton et al., 2024, Gao et al., 2024, Lieberum et al., 2024]. Recent work extends them beyond the residual stream to attention outputs [Kissane et al., 2024], while alternatives such as transcoders [Dunefsky et al., 2025] change the interpretability target and matching-pursuit SAEs [Costa et al., 2025] change the sparse inference procedure. Training variants such as gated SAEs, JumpReLU, and BatchTopK target the sparsity-fidelity tradeoff [Rajamanoharan et al., 2024a,b, Bussmann et al., 2024]. Absorption [Chanin et al., 2024], erasure-style evaluations [Karvonen et al., 2024], and SAEBench [Karvonen et al., 2025] argue that reconstruction error alone does not settle feature quality. Dense-latent analyses [Sun et al., 2025] likewise show that feature usefulness does not reduce to a single sparsity statistic. We follow that evaluation shift by treating downstream state replacement as a validation check rather than relying on MSE alone.

Structured decoders (bilinear, Kronecker, polynomial, multi-dimensional) impose factored structure on *vector* activations [Dooms and Gauderis, 2025, Kurochkin et al., 2025, Koromilas et al., 2026, Engels et al., 2024]; compact bilinear pooling [Gao et al., 2016] and Tucker-factored autoencoders use similar outer-product scoring. Our setting differs: the target is already matrix-valued and the rank-1 factorization follows from the model's own $\mathbf{k}_t\mathbf{v}_t^\top$ update rule, making the bilinear score a matched filter to the generative mechanism rather than a generic second-order lift. Structured decoders on non-vector targets (attention heads, state matrices) remain a nascent direction; this work is among the first to apply factored dictionaries to matrix-valued recurrent memory. The tied-bilinear ablation ($+7.05\%$ perplexity) confirms the encoder needs independent projections, distinguishing it from a symmetric matched filter.

The fast-weight view of linear attention [Schlag et al., 2021], formalized as test-time regression [Wang et al., 2025], led to GLA [Yang et al., 2023] and GatedDeltaNet [Yang et al., 2024], making matrix-valued recurrent memory a first-class architectural object rather than an implementation detail. Mechanistic studies now probe recurrent models via hidden-attention extraction [Ali et al., 2024], fact editing [Sharma et al., 2024], transfer analysis [Paulo et al., 2024], cross-architecture comparison [Wang et al., 2024, Arora et al., 2025], and activation subspace steering [Sunku Mohan et al., 2026]. Hossain et al. [2025] train autoencoders on Mamba hidden states without sparsity or structural constraints. We differ by matching the dictionary to the rank-1 write geometry, evaluating via downstream state replacement, and tracing the fidelity advantage to residual coherence across heads.

7 Discussion

Matching the encoder to the state's write geometry yields a consistent pattern across three models and three layers: lower downstream perplexity despite higher MSE. The mechanism is residual coherence: correlated flat errors compound under joint replacement, while bilinear errors stay independent (Section 5). Both the spectral predictor and the encoder advantage transfer to Qwen3.5-4B, where the rank-1 decoder penalty vanishes (-0.01% at 4B vs. $+11.18\%$ at 0.8B) as spectra concentrate with scale.

Interpretability. Write-vector alignment provides partial evidence for rank-1 structure discovery. Spearman probing against 57 text properties finds comparable per-feature interpretability for bilinear and flat (20% vs. 24%), but bilinear yields $3\times$ more alive features at matched training budget (1,400 vs. 400) and thus $3\times$ more total probe-detectable features. The strongest monovariant feature (bilinear, layer 9 head 4) tracks sentence-boundary position at Spearman $\rho{=}0.35$ ($p{<}10^{-106}$). Bilinear features also show partial write-vector alignment (17.5% above threshold; Section 4), but the mean effect remains modest (0.137 vs. 0.071 null), so this structural evidence constrains the mechanism without fully decomposing it.

Scope conditions. The bilinear advantage holds at $k \in \{16, 32\}$ and every tested n_f, but flat overtakes at $k{=}64$ and under BatchTopK sparsity ($+0.92\%$ flat vs. $+3.85\%$ bilinear), so the benefit is sparsity-regime-dependent. Mixed-position training (sampling states from all sequence positions, not just the final token) cuts flat's penalty ($+3.11 \rightarrow +\ 1.11\%$) but worsens bilinear ($+2.21 \rightarrow +\ 3.38\%$); the distributional shift interacts with architecture, and the direction depends on whether the encoder's geometry matches the position-conditioned state distribution. The spectral predictor transfers to DeltaNet but not to GLA's rectangular $(256{\times}512)$ states or Mamba-2 (Appendix I), limiting the current formulation to square state matrices.

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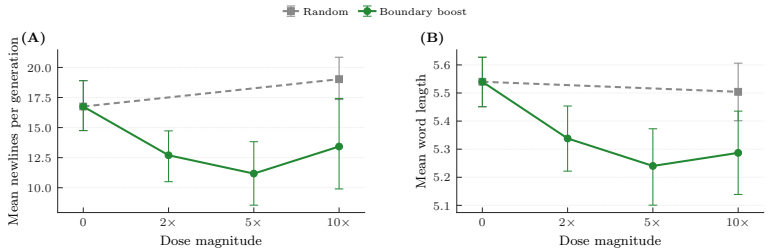

Figure 4: Causal steering on Qwen3.5-4B-Base, layer 9 (n=40 prompts, 400 tokens each). Boosting boundary-correlated BilinearSAE features reduces newlines dose-dependently up to 5×, with saturation at 10×. Random features produce no reduction. Word length shifts by ≤0.3 characters across all conditions.

A Causal Steering Probe

Protocol. We intervene on Qwen3.5-4B-Base at layer 9 with boundary-differential bilinear features across 40 prompts. Feature selection draws, from each of 32 heads, the 10 features with the largest sentence-boundary vs. non-boundary activation gap. Each prompt produces 400 tokens at temperature 0.7. At each step we add a positive offset to selected SAE coefficients before decoding the modified state back, preserving the residual. Three dose levels (2×, 5×, 10× the mean boundary activation) run against a matched-dose random-feature control. The primary readout is newline count; paragraphs and mean word length are secondary.

Results. Table 2 and Figure 4 show the dose-response. At 5× boost, mean newlines drop from 16.8 to 11.2, a 33% reduction (p=0.001, paired t-test; Cohen's d=0.55; n=40 prompts). The 2× dose also reaches significance (16.8→12.7, p=0.015). At 10×, the effect saturates and partially reverses (13.4 newlines), consistent with overdose destabilizing generation on a subset of prompts. Paragraphs track the same direction (7.5→6.2 at 5×); word length drops from 5.54 to 5.24, a 0.3-character shift. Random features at 10× dose produce 19.0 newlines, above baseline, consistent with generic perturbation adding noise to document structure.

Table 2: Steering metrics on Qwen3.5-4B-Base (95% bootstrap CIs, n=40 prompts, 400 tokens each). The primary comparison is 5× boost vs. baseline.

Condition	Dose	Newlines	Paragraphs	Word length
Baseline	0	16.8 [14.8, 18.9]	7.5 [6.6, 8.5]	5.54
Boost	2×	12.7 [10.1, 15.5]	6.3 [5.2, 7.4]	5.34
Boost	5×	11.2 [8.6, 13.9]	6.2 [4.9, 7.4]	5.24
Boost	10×	13.4 [10.1, 16.8]	7.4 [5.9, 8.9]	5.29
Random	10×	19.0 [17.2, 20.8]	7.8 [7.0, 8.7]	5.50

Controls. In a separate 0.8B experiment, FlatSAE boost collapses word length (4.86→3.53) while leaving paragraph count unchanged: the intervention degrades surface quality without shifting document structure. Only BilinearSAE features produced the target newline reduction. FlatSAE and MatrixSAE results come from separate 0.8B pilots with different feature pools and higher dead-feature rates (80% vs. 32%), so they serve as negative controls rather than matched comparisons. The boost/suppress asymmetry has a structural explanation: TopK activations are nonnegative, so suppression clamps coefficients to zero rather than inverting the feature direction. The sparse code has no negative loadings, limiting suppression to removing a feature's contribution at most, while boosting can amplify it without bound.

B k-Sweep Control

Table 3 reports the original layer-9 k-sweep from the first 5K extraction. We keep it because it isolates the effect of k within that cohort, but Appendix H is the canonical utilization audit across fresh subsets and larger training budgets.

Table 3: Dead feature count (out of 2048) across sparsity levels at n_f=2048, layer 9.

SAE	k=16	k=32	k=64
BilinearSAE	0 (0%)	0 (0%)	0 (0%)
MatrixSAE	231 (11%)	104 (5%)	0 (0%)
FlatSAE	1768 (86%)	1750 (85%)	1682 (82%)

FlatSAE stays above 80% dead regardless of k in this original sweep. BilinearSAE is at 0% dead for all three values, and MatrixSAE improves with higher k, reaching 0% dead at k=64. This within-sample control shows that stricter TopK sparsity worsens utilization, but it should not be read as a reproducible architecture ranking outside this extraction.

A separate BatchTopK sweep (dynamic per-batch threshold instead of fixed k) on fresh 5K subsets reduces dead features for all architectures, but BilinearSAE benefits least: dead rate drops from 93.3% to 83.1% (-10.2 pp) vs. FlatSAE's 86.6% to 73.0% (-13.6 pp). The bilinear encoder's structured scoring makes feature activation less threshold-sensitive, yielding diminishing returns from dynamic threshold adjustment.

C Per-Head Dead Feature Counts

Table 4 shows dead feature counts across all 16 GatedDeltaNet heads at layer 9, n_f=2048, k=32, from the clean per-head matched checkpoints used for Section 5. These are not the fresh-subset scaling runs from Appendix H; they summarize one matched-head training cohort.

Table 4: Average dead features (out of 2048) across 16 heads at layer 9, n_f=2048.

SAE	Mean dead	Min dead	Max dead	Std
BilinearSAE	656 (32%)	0	1532	505
MatrixSAE	1531 (75%)	0	1760	579
FlatSAE	1638 (80%)	0	1760	423

The utilization advantage of BilinearSAE is still visible after averaging across heads, but the head-to-head spread is large. Some heads are easy for every architecture, while others expose a sharp bilinear-versus-flat gap. This table supports the downstream cohort used in the main text, not the broader reproducibility claim.

D Full n_f-Sweep Data

Table 5 reports alive-feature percentages from the original 108-job sweep (3 seeds averaged). We retain it because it reveals how one extraction's architecture ordering depends on layer and dictionary size. Appendix H provides the canonical fresh-subset and budget-control results, which do not reproduce this ordering.

Table 5: Alive features (%) across dictionary sizes and layers. Each cell averages 3 seeds.

Layer	SAE	n_f=1024	n_f=2048	n_f=4096	n_f=8192
L1	BilinearSAE	100	100	100	100
	MatrixSAE	78	66	58	3
	FlatSAE	29	15	7	4
L9	BilinearSAE	100	100	100	100
	MatrixSAE	100	95	82	3
	FlatSAE	29	15	8	4
L17	BilinearSAE	100	91	65	4
	MatrixSAE	29	14	7	4
	FlatSAE	29	15	7	4

Within this original extraction, BilinearSAE remains highly alive through layers 1 and 9 while MatrixSAE tracks the spectral separation between layers 9 and 17. We therefore use this table as historical within-sample context, not as the final utilization claim.

E Feature Quality Metrics

Table 6 summarizes feature quality at n_f=2048, layer 9, for the head-0 feature-quality probe aggregated across matched k=32 checkpoints per SAE type.

In this matched k=32 slice, BilinearSAE keeps the largest alive set but distributes activity unevenly across it, yielding the highest Gini. MatrixSAE keeps the fewest alive features, but those features fire on larger fractions of the validation set and show the flattest usage among the alive subset.

Table 6: Feature quality metrics at n_f=2048, layer 9, averaged across matched head-0, k=32 checkpoints per SAE type.

SAE	Alive	Freq (%)	MSE	p_{95} MSE	Gini
BilinearSAE	170	21.6	1.32e-5	2.08e-5	0.761
MatrixSAE	72	44.5	1.33e-5	2.09e-5	0.461
FlatSAE	118	28.9	1.15e-5	1.83e-5	0.639

F Temporal Dynamics Categories

Table 7 reports the distribution of temporal categories across SAE architectures, aggregated across all checkpoints at layer 9.

Table 7: Temporal feature categories (% of alive features), averaged across the head-0 layer-9 checkpoints used in the temporal probe.

SAE	Late-emerging	Trans.-early	Transient	Persistent	Monotonic
BilinearSAE	33.0	23.5	15.7	8.2	1.3
MatrixSAE	32.3	21.6	17.0	8.9	0.8
FlatSAE	35.5	20.4	15.2	9.2	0.9

The maximum cross-architecture spread is 3.2 pp (late-emerging), consistent with temporal modes being properties of the GatedDeltaNet recurrence rather than the SAE architecture.

G Feature Probing Details

Text properties. For each corpus sequence, we compute 57 scalar properties from the decoded text, spanning six categories:
Surface statistics (15 properties): average word length, punctuation density, digit density, number count, uppercase ratio, newline density, quote density, code symbol density, URL presence, average sentence length, type-token ratio, whitespace density, character entropy, long word ratio, word count.
Syntactic proxies (10): function word ratio, content word ratio, comma density, semicolon/colon density, question mark density, exclamation density, max bracket nesting depth, open bracket count, word length standard deviation, sentence length standard deviation.
Information-theoretic (6): gzip compression ratio, word-level entropy, bigram repetition, trigram repetition, hapax legomena ratio (fraction of words appearing once), vocabulary growth rate (second-half vs. first-half unique words).
Structural/positional (6): last sentence boundary position, paragraph count, leading whitespace, line count, average line length, line length standard deviation.
Code indicators (6): programming keyword density (Python + JS), average indentation, max indentation, comment marker density, equals sign density, dot density.
Script and discourse (14): CJK character ratio, Cyrillic ratio, Latin ratio, non-ASCII ratio, dialogue line ratio, list line ratio, header density, pronoun density, first-person ratio, capitalized word ratio, average character ordinal, digit-letter transition density, consecutive repeat ratio, mean log word length.

Spearman correlation analysis. For each alive feature (activation frequency $\geq 1\%$), we compute Spearman rank correlation between its activation vector (across all N=2000 samples) and each property vector. A feature is flagged as interpretable when any property achieves $|\rho| \geq 0.15$ with Bonferroni-corrected p-value below $0.01/(n_{\text{alive}} \times 57)$. Spearman (rank-based) captures monotonic relationships without assuming linearity or normality in the zero-inflated activation distribution.

Nonlinear probing. Spearman correlation detects monotonic relationships but misses threshold effects, inverted-U patterns, and interactions between properties. For each alive feature, we train a random forest classifier (50 trees, max depth 6) to predict whether the feature's activation exceeds its median, using all 57 properties as inputs. Five-fold stratified cross-validation yields a mean accuracy per feature. A feature is flagged as nonlinearly interpretable if mean accuracy exceeds 60% (chance is 50%). Feature importances from the random forest identify which properties (and property combinations) drive the classification.

Contrastive probing. For each feature, we take the top-10% and bottom-10% sequences by activation magnitude (200 each). For each property, we compute Cohen's d (standardized mean difference) between these groups. A feature is flagged as contrastively interpretable when any property shows $|d| \geq 0.5$ with Bonferroni-corrected Mann-Whitney U $p < 0.01/57$. This method is more sensitive

than correlation for features with sharp activation thresholds: a feature that fires only on texts with digit density above 0.05 produces a large between-group d even if the rank correlation is modest.

Union interpretability. A feature counts as interpretable if *any* of the three methods flags it. The union captures features that encode linear trends (Spearman), nonlinear or combinatorial patterns (random forest), and threshold effects (contrastive). In the canonical layer-9 head-0 OpenWebText checkpoint used in the main paper, the union covers 52 of 196 alive features (26.5%). Across the four-head sweep, union counts range from 52 to 113 on OpenWebText and from 54 to 148 on UltraChat, so head content differs substantially even within one layer.

Structure vs. sparsity control. A TopK-disabled, ReLU-only rank-1 control sets $k = n_f$ at the same dictionary size ($n_f = 2048$). This leaves 719 features alive, of which 654 are interpretable (90.9%). The comparison to the sparse $k = 32$ run isolates the role of sparsity: the rank-1 parameterization already supports many probeable document-level directions, but TopK leaves most of them unused.

Held-out and corpus-shift checks. For head 0, the same-split OpenWebText run yields 52 interpretable features among 196 alive ones. On a held-out 20% split from that same 5K corpus, the union covers 34 of 36 alive features. On UltraChat, the same head yields 81 interpretable features among 171 alive ones on the same split and 34 of 35 on a held-out split. The held-out percentages are much larger because the alive set collapses to roughly 35 features in both corpora, so we use these runs as stability checks rather than as directly comparable interpretability rates.

Vocabulary projection. For rank-1 features, the decoder atom $\mathbf{v}_i\mathbf{w}_i^\top$ decomposes into a key direction $\mathbf{v}_i \in \mathbb{R}^{d_k}$ and value direction $\mathbf{w}_i \in \mathbb{R}^{d_v}$. We project $\mathbf{w}_i$ through the per-head slice of the output projection $W_O^{(h)} \in \mathbb{R}^{d_{\text{model}} \times d_v}$ and then through the unembedding matrix $W_U \in \mathbb{R}^{V \times d_{\text{model}}}$ to obtain a vocabulary distribution. Similarly, we project $\mathbf{v}_i$ through the transpose of the per-head query projection $W_Q^{(h)} \in \mathbb{R}^{d_k \times d_{\text{model}}}$ to find what input directions would query this feature. Top tokens from each projection indicate what the feature stores (via $\mathbf{w}$) and what contexts retrieve it (via $\mathbf{v}$).

Causal pilots. We tested two small causal pilots on five high-correlation sparse rank-1 features. Generation-based clamping to the 95th-percentile activation during 256-token continuations from 50 prompts produced 4 of 5 direction-aligned shifts, but no individual effect was significant and the sign test was not significant ($p = 0.188$). A higher-sensitivity direct-logit intervention on 200 prompts and four split positions per prompt also failed to clear the bar: 0 of 5 features reached $p < 0.05$, only 3 of 5 shifts were direction-aligned, and the mean absolute Cohen's d was 0.017. Follow-up head-12 ablations are cleaner but still modest. Single-feature ablations over 25 high-correlation features produce 17 of 25 directionally selective effects, but the mean selectivity is only +0.0001 and the strongest feature (word entropy) reaches +0.0007. Grouped property ablations are more consistent: 7 of 8 property families show positive max-dose selectivity, strongest for newline density (+0.0046), paragraph and line count (+0.0035), and word entropy (+0.0022). The grouped effects are consistent in sign, but even the largest stays in the 10^{-3} range.

H Data Scaling Analysis

This appendix documents the training-dynamics interaction behind utilization: the bilinear encoder recovers the lowest dead-feature rate given sufficient data (50K corpus, 200 epochs), but the advantage reverses on smaller budgets.

H.1 Subset reproducibility

We extract 50,000 sequences and partition them into five non-overlapping 5K subsets. Each subset trains all three architectures at n_f=2048, k=32, with 3 seeds per config (45 jobs per subset). Mean validation dead-feature percentages:

	FlatSAE	MatrixSAE	BilinearSAE
Mean dead %	86.6 ± 0.4	83.6 ± 0.8	93.3 ± 0.2

At 20 epochs on fresh 5K subsets, all three architectures have high dead-feature rates. The bilinear encoder's stronger inductive bias requires more optimization steps to activate its full dictionary, placing it behind the flat baseline at this training budget. The variance across subsets is small (<1 pp), so this ordering is stable within the 5K/20-epoch regime.

H.2 Training budget control

Training on the full 50K corpus with 200 epochs (10× the base schedule) restores the bilinear advantage:

	FlatSAE	MatrixSAE	BilinearSAE
Dead % (200 ep.)	93.4 ± 0.1	94.0 ± 0.4	85.8 ± 1.6

BilinearSAE recovers the lowest dead-feature rate with extended training. The matched-filter computation has a stronger inductive bias but fewer free parameters than the dense linear encoder, so the advantage emerges after more optimization. This slow-learner dynamic is consistent with the parameter-matched control in Section 5: a FlatSAE given the same parameter budget (8.4M) performs 5.5× worse on downstream perplexity regardless of training schedule.

H.3 Data diversity

The 5K and 50K corpora have nearly identical statistical properties: covariance effective rank 70.2 vs. 74.7 (ratio 1.06×), mean pairwise Frobenius distance 0.992 vs. 0.982 (ratio 0.99×), and identical σ_1/σ_2 ratios (5.998). These controls confirm that the ordering shift is a training-dynamics effect, not a data-diversity artifact. A BatchTopK control on the same 50K extraction lowers dead rates to 73.0% (FlatSAE), 73.3% (MatrixSAE), and 83.1% (BilinearSAE). Changing the sparsity rule helps but still leaves most features unused.

H.4 Reconstruction MSE scaling

Figure 5 shows validation MSE as a function of training corpus size. All runs use n_f=2048, k=32 (so L_0=32 throughout), with 3 seeds per configuration. FlatSAE achieves the lowest MSE at every corpus size from 5K onward. BilinearSAE starts highest at 1K samples but converges to within 15% of FlatSAE by 50K, consistent with the slow-learner dynamic. MatrixSAE tracks BilinearSAE at small corpora and converges between the other two at 50K.

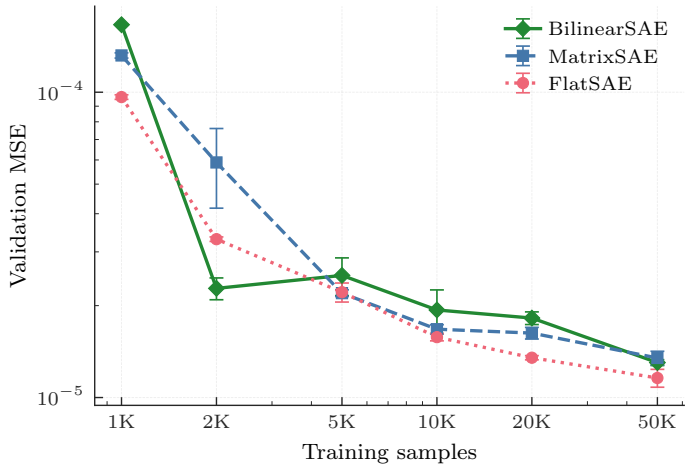

Figure 5: Validation MSE vs. training corpus size (layer 9, k=32, n_f=2048). Error bars show ±1 standard deviation across 3 seeds. All three architectures converge below 1.5×10^{-5} at 50K samples, with FlatSAE consistently lowest.

I Mamba-2 Spectral Audit

We run the same SVD diagnostic on Mamba-2 (AntonV/mamba2-780m-hf, 48 layers, 48 heads, state 64×128 per head) using 500 OpenWebText sequences. σ_1/σ_2 ranges from 1.39 (layer 12) to 1105.75 (layer 47), while effective rank ranges from 1.0 to 35.3 (mean 7.0); middle layers look distributed, while the last layer is nearly rank-1. This broad spread shows that wide spectral variation is not unique to the GatedDeltaNet setting. We treat it as a generalization check on the spectral diagnostic, not as evidence that the full SAE result already transfers to Mamba-2.

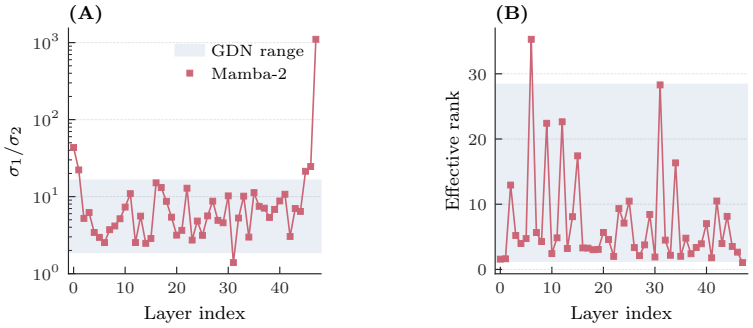

Figure 6: Mamba-2 SSM states across 48 layers (head 0, 500 sequences). **(A)** σ_1/σ_2 per layer (log scale); **(B)** effective rank per layer. Blue band: GDN range from Section 3. The wide range (1.4–1106) mirrors the GatedDeltaNet pattern.

J Corpus-Shift Robustness

Table 8 compares the matched layer-9 intervention on OpenWebText and UltraChat.

Table 8: Cross-corpus summary for the same-model robustness check.

Metric	OpenWebText	UltraChat
Layer-9 matched PPL delta, BilinearSAE	+1.33%	+1.54%
Layer-9 matched PPL delta, FlatSAE	+3.31%	+1.76%
Layer-9 matched PPL delta, MatrixSAE	+11.18%	+2.60%

UltraChat compresses the gaps: BilinearSAE (+1.54%), FlatSAE (+1.76%), MatrixSAE (+2.60%). All three fall within 1.1 percentage points on UltraChat, a tighter spread than the OpenWebText results. The matched layer-9 intervention survives a corpus shift.

K Full Encoder-Swap MSE Comparison

Table 9 reports layer-averaged validation MSE for all five encoder-decoder factorizations across three layers spanning concentrated to diffuse spectra. Flat SAEs achieve the lowest MSE on every layer. Bilinear-flat (bilinear encoder, dense decoder) ranks second on layers 1 and 9 but is beaten by rank-1 on layer 1. Tied bilinear has the highest MSE everywhere, consistent with its weight-sharing constraint.

Table 9: Layer-averaged validation MSE ($\times 10^{-5}$) across 16 heads, 3 seeds. Bold: lowest per layer. All five architectures use n_f=2048, k=32, 20 epochs.

	Layer 1 ($\sigma_1/\sigma_2 \approx 12$)	Layer 9 (6.8)	Layer 17 (3.0)
Flat (dense enc., dense dec.)	**2.61**	**2.58**	**30.1**
Rank-1 (dense enc., rank-1 dec.)	2.83	2.69	52.8
Bilinear (bilinear enc., rank-1 dec.)	4.07	3.09	54.0
Bilinear-flat (bilinear enc., dense dec.)	3.93	3.35	39.0
Tied bilinear (bilinear enc., tied rank-1)	7.63	5.98	80.5

The gap between flat and all structured variants widens at layer 17, where the spectral ratio is lowest (σ_1/σ_2=3.0). Rank-1 and bilinear have similar MSE on layer 17, both roughly 1.8× worse than flat, confirming the spectral diagnostic from Section 3: diffuse spectra penalize rank-1 constraints.

Parameter formulas. With d_k=d_v=128, d_{in}=$d_k d_v$=16,384: **FlatSAE** $= 2n_f d_{\text{in}} + n_f + d_{\text{in}}$ (encoder W_e, decoder W_d, biases); **MatrixSAE** $= n_f d_{\text{in}} + n_f(d_k + d_v) + n_f + d_k d_v$ (dense encoder, rank-1 decoder V+W, biases); **BilinearSAE** $= 4n_f d_k + n_f + d_k d_v$ (encoder V_{enc}+W_{enc}, decoder V_{dec}+W_{dec}, biases).

L Hyperparameter Fairness Details

This appendix documents training and inference choices that could otherwise bias the encoder-swap comparison.

Encoder factor normalization. For BilinearSAE, we unit-normalize each encoder factor pair $(\mathbf{v}_i, \mathbf{w}_i)$ after every optimizer step, so the matched-filter score $\mathbf{v}_i^\top S_t \, \mathbf{w}_i$ is scale-invariant across atoms. Without this normalization, atoms with larger-norm factors win TopK selection through

scale rather than alignment. FlatSAE and MatrixSAE apply the standard decoder-column re-normalization every 100 steps used throughout the SAE literature [Cunningham et al., 2023, Bricken et al., 2023]; bilinear atoms add the analogous encoder-factor re-normalization.

TopK at training vs. inference. At training and at inference time, TopK selects the 32 highest-scoring atoms per input (k=32 throughout the matched cohort). Scores are the pre-activation encoder outputs (flat: $W_e \, \mathrm{vec}(S_t)$; bilinear: $\mathbf{v}_i^\top S_t \, \mathbf{w}_i$), before nonnegativity is enforced. Nonnegativity is applied by clamping the selected 32 coefficients at zero: negative scores survive TopK selection if their magnitudes are among the 32 largest, but their decoded contribution is zeroed after clamping. This matches the TopK SAE convention in Gao et al. [2024].

Bias handling. No encoder or decoder bias terms are used for FlatSAE, MatrixSAE, or BilinearSAE. Mean-centering uses the per-head empirical mean M_h computed from the training split (frozen across seeds); $\mathbf{x}$=$\mathrm{vec}(S_t - M_h)$ is the SAE input for flat and rank-1, and $S_t - M_h$ for bilinear. Means are held fixed at inference; we do not recompute M_h on the evaluation corpus.

Optimizer parity. All five encoder-swap variants share the same optimizer (Adam), cosine schedule (50-step warmup, peak $3{\times}10^{-4}$, decay to $3{\times}10^{-5}$), batch size (256), training length (20 epochs), and dead-feature resampling cadence (every 250 steps for features inactive ≥ 100 consecutive steps). Only the factorization differs.